

ORIGINAL ARTICLE

Hemodynamics and Renin Profiles of Renal Artery Variations in Hypertensive CKD

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BACKGROUND: The functional implications of renal artery variations—including accessory renal artery (ARA) and early branching—in hypertensive chronic kidney disease remain poorly understood.

METHODS: This prospective cohort study enrolled 38 patients with hypertensive chronic kidney disease and unilateral solitary renal artery variation (24 ARA, 14 early branching) undergoing pressure wire–based hemodynamic evaluation and selective segmental renal vein renin sampling; 35 patients completed 2-year follow-up.

RESULTS: Mean pressure gradient was higher in variant artery than in ipsilateral main (1.10 ± 3.39 versus -0.15 ± 2.21 mmHg, $P=0.04$) and contralateral renal artery (versus -0.10 ± 1.56 mmHg, $P=0.04$) in the overall cohort, a difference limited to the ARA subgroup. Plasma renin activity was elevated in ARA-supplied segments compared with the ipsilateral main trunk (1.77 ± 1.86 versus 1.24 ± 1.40 ng/(mL·h), $P=0.004$) and contralateral artery–supplied segments (versus 1.24 ± 1.37 , $P=0.002$), whereas no significant difference was observed in the early branching subgroup. After multivariable adjustment, variant artery systolic pressure gradient was positively associated with variant-supplied segmental ($\beta=0.41$, $P=0.02$) and peripheral ($\beta=0.37$, $P=0.04$) plasma renin activity, angiotensin II, and nighttime mean arterial pressure. Peripheral aldosterone was associated with lower estimated glomerular filtration rate and higher urinary protein. All adjusted associations represent exploratory signals given limited sample size. Over 2 years, an intragroup estimated glomerular filtration rate decline was observed within the ARA subgroup (-5.2 mL/[min·1.73 m²]; $P=0.02$) while the early branching subgroup remained stable ($P=0.47$); however, the group-by-time interaction was not significant ($P=0.06$), indicating these preliminary trends require larger validation cohorts.

CONCLUSIONS: Renal artery variations may alter intrarenal hemodynamics, with functional impairment and segmental renin hypersecretion observed primarily in ARA. This suggests a potential role for functional evaluation. (*Hypertension*. 2026;83:2874–2887. DOI: 10.1161/HYPERTENSIONAHA.126.27130.) • **Supplement Material.**

Key Words: hemodynamics ■ hypertension ■ renal artery ■ renal insufficiency, chronic ■ renin

Chronic kidney disease (CKD) complicated with hypertension represents a major global public health challenge.^{1,2} Hypertension is both a leading cause and a major driver of progressive renal function deterioration in CKD.³ Renal artery variations—primarily accessory renal artery (ARA) and early branching (EB)—affect ≈20% to 35% of the general population, with even higher prevalence in patients with hypertension.^{4,5} Renal artery variations occur due to incomplete regression of embryonic

renal arteries during development.⁶ ARAs are renal arteries that arise from the abdominal aorta or its branches. EB refers to an arterial bifurcation located within 20 mm from the origin of the renal artery trunk.^{4,7,8}

In patients with hypertension and CKD, the clinical significance of renal artery variations has not been fully elucidated. In 1951, Marshall⁹ first proposed a link between renal artery variations and hypertension, although subsequent studies yielded

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NOVELTY AND RELEVANCE

What Is New?

In patients with hypertension and chronic kidney disease, accessory renal arteries—but not early branching—exhibit distinct hemodynamic alterations, with significantly elevated mean and diastolic pressure gradients compared with ipsilateral main and contralateral arteries. Renin hypersecretion is observed localized to accessory renal artery-supplied territories, and the variant-induced systolic pressure gradient is associated with both localized and systemic renin activation.

What Is Relevant?

Renal artery variations, particularly accessory renal arteries, may represent functionally active subclinical stressors that alter intrarenal hemodynamics in

hypertensive chronic kidney disease, rather than benign anatomic curiosities. Pressure wire-derived hemodynamic assessment combined with selective segmental renal vein renin sampling provides a novel, physiology-based approach to explore the functional significance of renal artery variations.

Clinical/Pathophysiological Implications?

These proof-of-concept findings highlight a potential transition from traditional anatomic classification to functional, physiology-based evaluation of renal artery variations, offering a potential framework that carries future clinical implications for refining risk stratification and optimizing cardiorenal protection in this specific phenotype.

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
ARA	accessory renal artery
BP	blood pressure
DPG	diastolic pressure gradient
EB	early branching
IVST	interventricular septal thickness
LVPWT	left ventricular posterior wall thickness
MPG	mean pressure gradient
PG	pressure gradient
PRA	plasma renin activity
PRA_{sm}	mean segmental renal vein plasma renin activity
PRA_{sv}	plasma renin activity from renal segments supplied by variant renal artery
PRA_t	plasma renin activity from renal vein trunk
RAAS	renin-angiotensin-aldosterone system
SPG	systolic pressure gradient
UACR	urinary albumin-to-creatinine ratio

inconsistent results.^{5,10} Recent clinical studies have suggested that ARA may contribute to the development of hypertension and target-organ damage,^{11–13} with the mechanism possibly linked to overactivation of the renin-angiotensin-aldosterone system (RAAS) and sympathetic nervous system.¹⁴ Moreover, ARAs reduced the effect of blood pressure (BP) reduction after renal denervation.¹⁵ However, whether these anatomic variants cause functional hemodynamic disturbances, trigger localized RAAS activation, and contribute to hypertension and CKD progression remains

unclear due to the lack of standardized functional assessments. Resolving this bottleneck is critical in this high-risk population to identify occult secondary hypertension mechanisms and target-organ damage, strongly justifying invasive functional evaluation for optimized clinical management.

Pressure wire-based assessment has become the gold standard for evaluating hemodynamic significance in renal artery stenosis.¹⁶ By measuring translesional pressure gradient (PG), this technique guides revascularization decisions and facilitates the transition from anatomic diagnosis to functional assessment.¹⁷ Furthermore, selective renal vein renin sampling is a classic method for identifying intrarenal RAAS hyperactivation.^{18–20} The combination of the 2 methods provides a comprehensive and objective functional assessment of renal artery variations, thereby facilitating investigations into their pathophysiological significance in hypertension and CKD.

To the best of our knowledge, this is the first study using pressure wire assessment to evaluate the hemodynamic significance of renal artery variations in hypertensive CKD. Combining pressure wire-derived hemodynamics with selective renal vein renin activity, we hypothesized that variant-induced perfusion changes trigger local RAAS activation and distinct functional impairments. This study aims to provide evidence for the functional evaluation and individualized therapy of renal artery variations in hypertensive CKD.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Design and Participants

This prospective cohort study enrolled patients with hypertensive CKD and unilateral solitary renal artery variation (ARA or EB) from Ruijin Hospital affiliated with Shanghai Jiao Tong University School of Medicine between January 2021 and December 2023. The inclusion criteria were patients with hypertension (BP $\geq 140/90$ mmHg) aged 18 to 60 years with an estimated glomerular filtration rate (eGFR) of 15 to 59 mL/(min $\cdot$ 1.73 m 2) or urinary albumin-to-creatinine ratio (UACR) of 3 to 200 mg/mmol for at least 3 months, and suspected unilateral solitary renal artery variation identified on noncontrast-enhanced magnetic resonance angiography and Doppler ultrasound. Exclusion criteria included renal artery stenosis, bilateral or multiple renal artery variations; diabetes, endocrine hypertension, autoimmune disease, malignancies, pregnancy, contrast allergy, recent (≤ 4 weeks) glucocorticoid use, or prior renal or vascular interventions affecting renal hemodynamics (eg, angioplasty, stenting, or partial nephrectomy). Given the invasive nature of pressure wire assessments and selective sampling, this was designed as a pilot mechanistic study. Sample size was dictated by consecutive enrollment over 3 years rather than a formal power calculation. An intraindividual design was utilized to minimize inter-subject variability and control for systemic confounders. Antihypertensive medications were adjusted for all enrolled patients to guarantee the accuracy

of plasma renin activity (PRA) measurements in accordance with the Endocrine Society Clinical Practice Guideline.²¹ Diuretics and aldosterone antagonists were withdrawn for at least 4 weeks; ACE (angiotensin-converting enzyme) inhibitors, angiotensin II receptor blockers, angiotensin receptor–neprilysin inhibitor, β -adrenergic blockers, dihydropyridine calcium channel blockers, and clonidine were discontinued for at least 2 weeks. Instead, nondihydropyridine calcium channel blockers and α -adrenergic blockers were administered for BP control. This washout duration balanced methodological rigor with patient safety, since prolonged RAAS inhibitor discontinuation increases mortality and renal deterioration risk in patients with CKD.²² The study protocol was approved by the institutional ethics committee (Approval No. 2020-319) and registered at the Chinese Clinical Trial Registry (URL: <https://www.chictr.org.cn/showproj.html?proj=64623>; Unique identifier: ChiCTR2000040683). All participants provided written informed consent. The study flowchart is presented in Figure 1.

Pressure Wire–Based Hemodynamic Assessment

Under local anesthesia, femoral arterial access was obtained. Percutaneous renal artery angiography was performed to

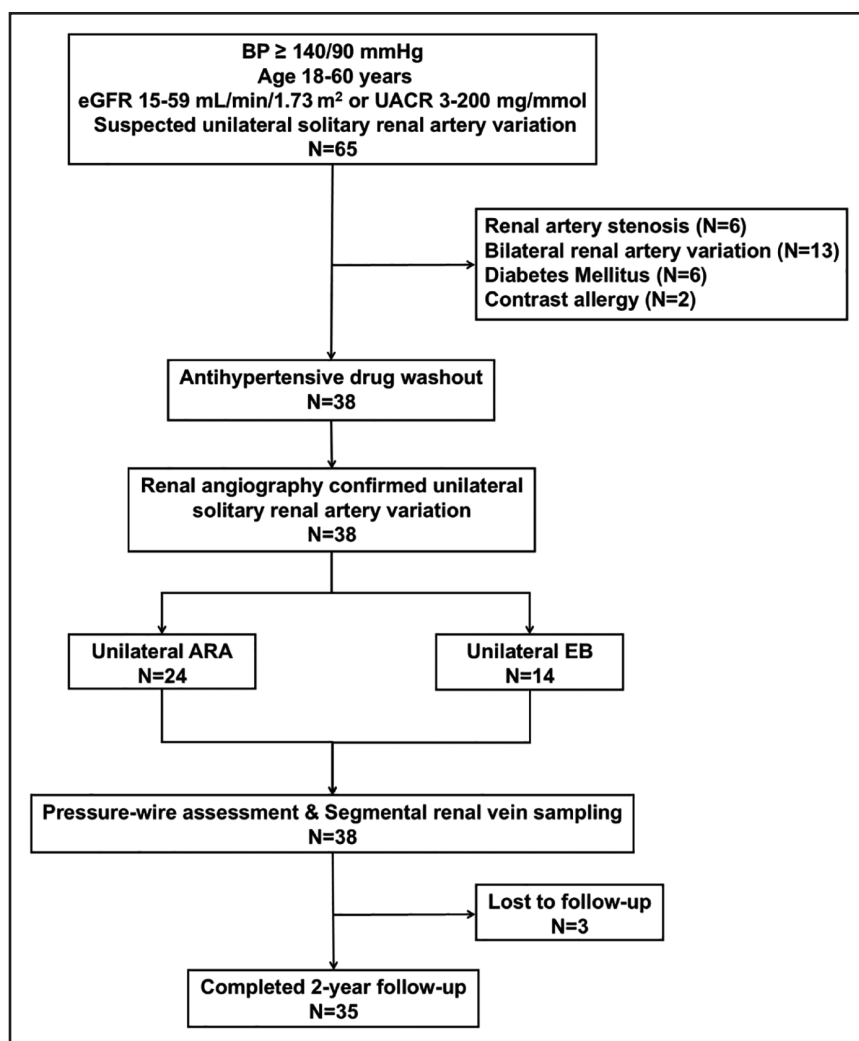


Figure 1. Flowchart of the study population.

ARA indicates accessory renal artery; BP, blood pressure; EB, early branching; eGFR, estimated glomerular filtration rate; and UACR, urinary albumin-to-creatinine ratio.

confirm a unilateral solitary anatomic variant by 2 skilled interventional radiologists. A 6F guiding catheter was placed in the opening of the renal artery, and a 0.014-inch pressure wire (St. Jude Medical, St. Paul, Minnesota) was advanced into the renal artery. After pressure equalization, systolic, diastolic, and mean arterial pressures (MAP) were recorded at 3 locations (ostium, mid-segment, and distal segment adjacent to the renal hilum) within each of the 3 arteries (variant, ipsilateral main trunk, and contralateral renal artery). The mean value of these 3 measurements was used to represent the aortic pressure for each artery. Subsequently, segmental renal artery pressures were measured by advancing the pressure wire to the midpoint of each segmental renal artery. The mean value of these segmental pressure measurements was used to represent segmental renal artery pressure. The PG was calculated as the difference between the variant or main renal artery pressure and the corresponding segmental renal artery pressure (Figure 2). Hemodynamic assessments followed a strict protocol: transducer calibration and pressure equalization were performed at the renal artery ostium before wire advancement. Pressures were recorded on waveform stabilization across 3 to 5 cardiac cycles during quiet breathing, followed by uniform pull-back. All

procedures utilized identical wire models and equipment, performed by a consistent operator-technician team. Independent investigators conducted offline analysis.

Selective Segmental Renal Vein Renin Sampling

Via femoral venous access, selective catheterization of renal vein branches was performed using a 5F Simon catheter. Under fluoroscopic guidance, blood samples were obtained from the renal vein trunks and respective segmental branches (typically 2–3 each) of both the variant and contralateral sides. The inferior vena cava sample was collected at a level 1 vertebra above the renal vein ostium as a peripheral reference. PRA was measured by radioimmunoassay. For accurate PRA testing, blood samples were processed via strict cold-chain protocols and refrigerated centrifugation. Renal vein trunk PRA was denoted as PRA from renal vein trunk (PRA_v). Given that renin secretion is pulsatile, the mean value of segmental renal vein PRA was calculated and denoted as mean segmental renal vein PRA (PRA_{sm}), where $PRA_{sm} = (PRA_{segment1} + PRA_{segment2} + \dots + PRA_{segmentn}) / n$ (typically $n=2-3$ branches). For the anatomically variant side,

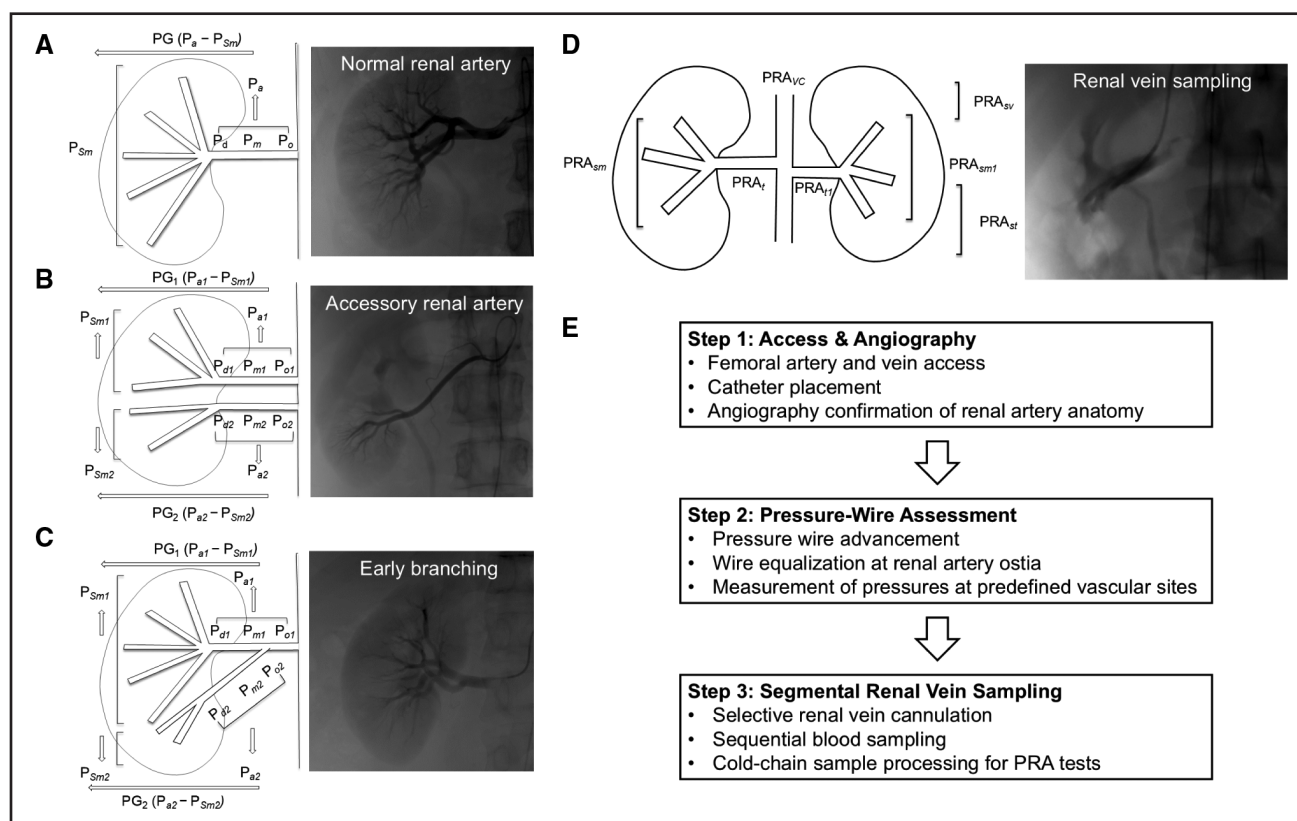


Figure 2. Schematic diagrams and procedural workflow for comprehensive functional assessment.

A through **C**, Schematic diagram of pressure wire–based renal artery pressure measurement in different anatomic phenotypes. P_a represents aortic pressure, defined as the mean pressure measured by the pressure wire at the ostium (P_o), mid-segment (P_m), and distal segment (P_d) of the renal artery main trunk. Mean segmental renal artery pressure (P_{sm}) is calculated as the average pressure measured in each segmental renal artery using the pressure wire. The pressure gradient (PG) was calculated by subtracting the corresponding P_{sm} from the P_a . Subscript 1 denotes the ipsilateral main renal artery, and 2 denotes the anatomically variant renal artery. **D**, Schematic diagram of selective segmental renal vein renin sampling. Plasma renin activity (PRA) from renal vein trunk (PRA_v), mean segmental renal vein PRA (PRA_{sm}), PRA from renal segments supplied by variant renal artery (PRA_{sv}), PRA from renal segments supplied by main trunk renal artery (PRA_{st}), and PRA from vena cava (PRA_{vc}) denote PRA at respective sampling sites, with subscript 1 designating the anatomically abnormal side. **E**, Procedural workflow diagram.

segmental renal vein PRA was distinguished according to the renal segments supplied by the variant renal artery (PRA from renal segments supplied by variant renal artery [PRA_{sv}]) versus the PRA from renal segments supplied by the main trunk renal artery, based on their respective drainage territories, to facilitate subsequent analysis (Figure 2).

Clinical and Biochemical Assessments

Baseline clinical data were collected for all enrolled patients, including demographics, hypertension characteristics (duration, medication history), biochemical parameters (blood glucose, liver function, lipid profile, electrolytes), renal biomarkers (serum creatinine, urinary protein quantification, UACR), cardiac biomarkers (troponin I, NT-proBNP [N-terminal pro-B-type natriuretic peptide]), and echocardiographic parameters. The Chronic Kidney Disease Epidemiology Collaboration equation was used to calculate eGFR. Left ventricular mass was calculated using the Devereux formula and indexed to body surface area to derive the left ventricular mass index. Basal PRA, plasma aldosterone, and angiotensin II levels were measured after 8 hours of overnight supine rest, whereas stimulated levels were measured after 2 hours of upright standing. 24-hour ambulatory BP monitoring was performed using an Ambulatory BP Monitor (SPACELABS 90217).

Standardized RAAS-Targeted Therapy and 2-Year Follow-up Protocol

After hemodynamic assessment and renal vein sampling procedure, all enrolled patients received standardized antihypertensive therapy with RAAS inhibition as the cornerstone, including either ACE inhibitors, angiotensin II receptor blockers, or angiotensin receptor–neprilysin inhibitor. The specific agent was selected at the treating physician's discretion based on tolerability and comorbidities, with dose titration to maximally tolerated doses within 3 months of enrollment. Additional antihypertensive agents (calcium channel blockers, β -adrenergic blockers, and diuretics) were added sequentially to achieve target office BP <130/80 mm Hg.

Participants were prospectively followed for 24 months, with structured evaluations performed every 6 months. At each visit, the following parameters were systematically assessed: (1) office BP (mean of 3 measurements after a 5-minute rest); (2) antihypertensive medication regimen (drug classes, doses, and adherence); (3) serum creatinine for eGFR calculation (Chronic Kidney Disease Epidemiology Collaboration equation) and UACR.

The primary follow-up outcomes were as follows: (1) the time-averaged office BP over the 2 years; (2) the eGFR decline over 2-year follow-up; and (3) progression of UACR, defined as a >50% increase from baseline or progression to >26.5 mg/mmol. Given the 2-year duration and anticipated low hard end point event rate, we prespecified these continuous trajectories as surrogate markers of renal outcomes, acknowledging that longer follow-up would be required for definitive clinical outcome assessment.

Statistical Analysis

Continuous variables were expressed as mean $\pm$ SD or median (range) according to the Shapiro-Wilk test. Categorical

variables were reported as n (%) and compared using the χ^2 test or Fisher exact test when expected frequencies were <5. Paired Student *t* test or Wilcoxon signed-rank test was used for within-subject comparisons, and unpaired Student *t* test or Mann-Whitney *U* test for between-group comparisons. Longitudinal eGFR trajectories were evaluated using linear mixed-effects models fitted via restricted maximum likelihood. The primary model incorporated subject-specific random intercepts and random time slopes (unstructured covariance matrix). Fixed effects included group, continuous time (in years), and group-by-time interaction to assess slope differences. Under the missing-at-random assumption, all 38 participants with at least 1 valid eGFR measurement were retained without missing value imputation. Point-wise least-squares means for visualization were derived from an auxiliary model with categorical time, maintaining the same random-effects structure. Correlations between hemodynamic parameters, PRA, cardiorenal biomarkers, and 2-year changes in eGFR and UACR were analyzed using Pearson (normal distribution) or Spearman (non-normal distribution) correlation. To account for potential confounding, multivariable linear regression was performed to evaluate adjusted associations. All baseline hemodynamic and biochemical data were complete with no missing values. Statistical analyses were performed using SPSS 26.0 and R 4.6.1. Statistical significance was defined as $P < 0.05$.

RESULTS

Baseline Characteristics

Between January 2021 and December 2023, a total of 38 patients with hypertensive CKD and unilateral solitary renal artery variation were enrolled (24 ARA and 14 EB; Figure 1). Baseline characteristics are presented in Table 1. The average age was 40 $\pm$ 11 years, and 29 (76%) were male. The average age of hypertension onset was 35 $\pm$ 10 years, whereas the median duration of hypertension was 1 year. Fifteen (40%) patients experienced at least 1 episode of severe hypertension (BP $\geq$ 180/120 mm Hg). The patients with EB had a higher body mass index than those with ARA. No significant differences were observed between the 2 subgroups in terms of age, sex, fasting plasma glucose, hepatic function, lipid profile, serum and urinary sodium and potassium levels, as well as basal and stimulated peripheral renin, angiotensin II, and aldosterone levels. The antihypertensive medications and BP control status before angiography were comparable between the 2 subgroups. Furthermore, between-group differences were not significant for renal biomarkers or cardiac biomarkers.

Hemodynamic Assessment by Pressure Wire

Pressure wire–derived hemodynamic parameters were measured in all enrolled patients (Figure 3; Figure S1). Mean PG (MPG) of the variant renal artery (1.10 $\pm$ 3.39 mm Hg) was higher than that of both the ipsilateral

Table 1. Demographic, Biochemical, Target-Organ, and BP Control Characteristics of the Study Population at Baseline

	Total (N=38)	ARA (n=24)	EB (n=14)	P value
Demographics				
Age, y	40±11	38±11	44±11	0.17
Male, n (%)	29 (76)	19 (79)	10 (71)	0.70
BMI, kg/m ²	24.1±4.1	23.0±2.4	26.1±5.5	0.02
Smoker, n (%)	6 (16)	5 (21)	1 (7)	0.38
Age of hypertension onset, y	35±10	34±11	36±9	0.63
Hypertension duration, y	1 (0.1–20)	1 (0.1–20)	6 (0.1–22)	0.14
Episode of severe hypertension, n (%)	15 (40)	9 (38)	6 (43)	0.74
Family history of hypertension, n (%)	11 (29)	7 (29)	4 (29)	1.00
Medication before withdrawal				
ACE inhibitors/ARB/ARNI, n (%)	28 (73.7)	18 (75.0)	10 (71.4)	0.81
DCCB, n (%)	17 (44.7)	13 (54.2)	4 (28.6)	0.13
BB, n (%)	8 (21.1)	6 (25.0)	2 (2.9)	0.68
AB, n (%)	4 (10.5)	1 (4.2)	3 (21.4)	0.13
Diuretic, n (%)	5 (13.2)	4 (16.7)	1 (7.1)	0.63
No. of antihypertensive drug	2 (1–5)	2 (1–5)	1 (1–4)	0.27
Medication after withdrawal				
NDCCB, n (%)	30 (78.9)	20 (83.3)	10 (11.1)	0.43
AB, n (%)	25 (65.8)	15 (62.5)	10 (71.4)	0.73
Biochemical variables				
FPG, mmol/L	4.53±0.55	4.54±0.53	4.52±0.59	0.96
ALT, U/L	18±6	18±6	18±5	0.92
AST, U/L	18±3	19±4	18±3	0.58
Triglycerides, mmol/L	1.76±1.72	1.52±0.82	2.18±2.64	0.26
TC, mmol/L	4.50±0.93	4.34±0.85	4.77±1.03	0.17
HDL-C, mmol/L	1.18±0.30	1.14±0.28	1.25±0.32	0.27
LDL-C, mmol/L	2.73±0.78	2.63±0.67	2.91±0.94	0.29
Lipoprotein(a), mmol/L	0.34±0.35	0.31±0.34	0.38±0.39	0.54
Serum sodium, mmol/L	142.0±1.8	142.2±2.0	141.7±1.3	0.46
Serum potassium, mmol/L	3.73±0.35	3.72±0.38	3.75±0.31	0.76
Urinary sodium, mmol/24 h	142.16±70.59	139.96±72.65	145.94±69.43	0.80
Urinary potassium, mmol/24 h	33.44±13.54	32.17±12.77	35.61±15.01	0.46
Basal renin activity, ng/(mL·h)	1.66±1.21	1.90±1.33	1.23±0.87	0.10
Stimulated renin activity, ng/(mL·h)	3.85±2.38	4.18±2.42	3.28±2.26	0.27
Basal angiotensin II, pg/mL	65.00±12.60	66.40±13.40	62.58±11.13	0.38
Stimulated angiotensin II, pg/mL	79.30 (43.66–169.56)	85.94 (43.66–169.56)	75.36 (63.24–158.29)	0.58
Basal aldosterone, pg/mL	90.10±46.94	92.86±44.22	85.32±52.65	0.64
Stimulated aldosterone, pg/mL	207.31 (39.35–719.16)	195.34 (39.35–719.16)	220.60 (91.68–619.45)	0.56
Renal damage				
SCr, μmol/L	106±33	102±25	114±44	0.30
eGFR, mL/(min·1.73m ²)	77±23	81±24	70±19	0.15
Urinary protein, mg/24 h	278 (48–2996)	300 (48–2996)	238 (82–2775)	0.40
UACR, mg/mmol	10.60 (0.41–194.06)	11.63 (0.66–194.06)	7.08 (0.41–94.47)	0.33
Cardiac damage				
NT-proBNP, pg/mL	49.7 (5.0–1864.0)	45.0 (5.0–339.7)	84.5 (10.0–1864.0)	0.24
Troponin I, ng/mL	0.01 (0.01–0.04)	0.01 (0.01–0.04)	0.01 (0.01–0.04)	0.62

(Continued)

Table 1. Continued

	Total (N=38)	ARA (n=24)	EB (n=14)	P value
LVEDD, mm	48 (29–63)	48 (41–55)	48 (29–63)	0.75
IVST, mm	9 (7–13)	9 (7–13)	9 (8–13)	0.13
LVPWT, mm	9 (7–13)	9 (7–13)	9 (8–13)	0.20
LVMI, g/m ²	77 (45–198)	77 (56–150)	79 (45–198)	0.63
LVEF, %	67±6	68±3	65±8	0.18
BP control status				
24 h SBP, mm Hg	125±14	127±13	124±15	0.58
24 h DBP, mm Hg	80±10	81±11	80±9	0.86
24 h MAP, mm Hg	95±11	95±11	94±11	0.81
Daytime SBP, mm Hg	127±14	128±12	125±16	0.54
Daytime DBP, mm Hg	81±10	82±11	81±10	0.85
Daytime MAP, mm Hg	96±11	96±10	95±11	0.80
Nighttime SBP, mm Hg	121±16	121±16	120±15	0.91
Nighttime DBP, mm Hg	76±11	76±12	77±9	0.85
Nighttime MAP, mm Hg	90±14	91±13	87±17	0.49
Nighttime dipping score, %	4.7±7.8	5.5±7.9	3.5±7.7	0.45
Nighttime dippers, n (%)	11 (28.9)	5 (20.8)	6 (42.9)	0.27
Resistant hypertension, n (%)	1 (3)	1 (4)	0 (0)	1.00

Nighttime dippers were defined as subjects with a $\geq 10\%$ decline in mean SBP from daytime to nighttime. Resistant hypertension was defined as uncontrolled BP ($\geq 140/90$ mmHg) despite adherence to a maximally tolerated regimen of ≥ 3 antihypertensive agents (including a diuretic), or controlled BP requiring ≥ 4 agents. Data are expressed as mean $\pm$ SD, median (range), or n (%). AB indicates α -Adrenergic blocker; ACE, angiotensin-converting enzyme; ALT, alanine transaminase; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; AST, aspartate transaminase; BB, β -Adrenergic blocker; BMI, body mass index; DBP, diastolic blood pressure; DCCB, dihydropyridine calcium channel blocker; eGFR, estimated glomerular filtration rate; FPG, fasting plasma glucose; HDL-C, high-density lipoprotein cholesterol; IVST, inter-ventricular septal thickness; LDL-C, low-density lipoprotein cholesterol; LVEDD, left ventricular end-diastolic diameter; LVEF, left ventricular ejection fraction; LVMI, left ventricular mass index; LVPWT, left ventricular posterior wall thickness; MAP, mean arterial pressure; NDCCB, nondihydropyridine calcium channel blocker; NT-proBNP, N-terminal pro-B-type natriuretic peptide; SBP, systolic blood pressure; SCr, serum creatinine; TC, total cholesterol; and UACR, urinary albumin-to-creatinine ratio.

main renal artery (-0.15 ± 2.21 mmHg, $P=0.04$) and the contralateral renal artery (-0.10 ± 1.56 mmHg, $P=0.04$). Both systolic PG (SPG) and diastolic PG (DPG) did not differ among the variant artery, ipsilateral artery, and contralateral artery. In the subgroup analysis, hemodynamic abnormalities were isolated to ARA, but not EB. In patients with ARA, both MPG and DPG of the ARA (MPG, 1.67 ± 2.72 mmHg; DPG, 1.51 ± 3.08 mmHg) were higher than those of the ipsilateral main (MPG, -0.40 ± 2.46 mmHg, $P=0.006$; DPG, -0.74 ± 3.33 mmHg, $P=0.02$) and the contralateral renal artery (MPG, 0.03 ± 1.64 mmHg, $P=0.01$; DPG, -0.60 ± 2.46 mmHg, $P=0.01$), whereas SPG did not differ among the 3 arteries. In patients with EB, MPG, DPG, and SPG showed no significant differences across these arteries.

Renal Vein Renin Profile in Patients With Hypertensive CKD and Unilateral Solitary Renal Artery Variation

Selective renal vein renin sampling was performed in all study participants. Renal vein PRA differed significantly by sampling location (Figure 4). PRA_{sm} was significantly higher on the variant side compared with the

contralateral normal side (1.56 ± 1.45 versus 1.36 ± 1.26 ng/(mL·h), $P=0.04$), whereas no significant difference was observed in PRA_t between sides (1.59 ± 1.44 versus 1.52 ± 1.49 ng/(mL·h), $P=0.67$), indicating localized RAAS activation masked at the composite level. Within the ARA subgroup, PRA_{sm} exhibited marked lateralization (1.57 ± 1.62 versus 1.24 ± 1.37 , $P=0.008$), whereas PRA_t showed no significant between-side disparity (1.78 ± 1.68 versus 1.53 ± 1.70 ng/(mL·h), $P=0.32$). Conversely, no such between-side differences were observed in the EB subgroup for either PRA_{sm} (1.55 ± 1.18 versus 1.58 ± 1.04 ng/(mL·h), $P=0.81$) or PRA_t (1.26 ± 0.86 versus 1.49 ± 1.10 ng/(mL·h), $P=0.24$). Notably, PRA_{sv} was markedly higher than PRA from renal segments supplied by the main trunk renal artery (1.82 ± 1.78 versus 1.36 ± 1.26 ng/(mL·h), $P=0.001$), and also exceeded the PRA_{sm} on the contralateral side ($P=0.003$). Subgroup analysis demonstrated that PRA_{sv} was significantly higher than PRA from renal segments supplied by the main trunk renal artery in the ARA subgroup (1.77 ± 1.86 versus 1.24 ± 1.40 ng/(mL·h), $P=0.004$) with corresponding elevation in contralateral comparison ($P=0.002$), whereas the EB subgroup showed no significant differences either between vascular territories or between kidneys.

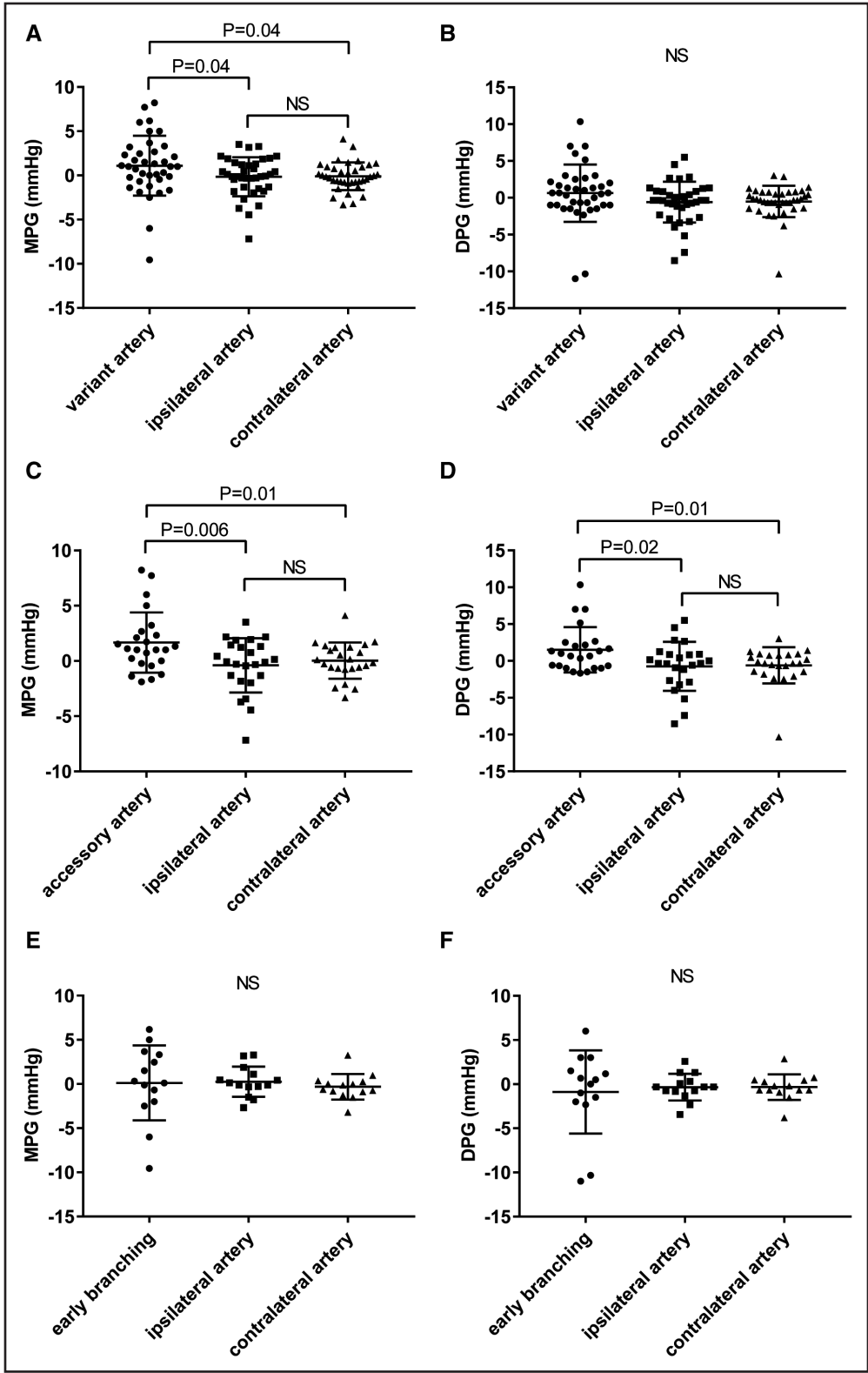


Figure 3. Pressure gradient comparisons across variant, ipsilateral main, and contralateral renal artery in patients with unilateral solitary renal artery variation. Comparisons of mean pressure gradient (MPG) and diastolic pressure gradient (DPG) among the variant, ipsilateral main, and contralateral renal artery are presented for the overall cohort (A and B), the accessory renal artery subgroup (C and D), and the early branching subgroup (E and F), respectively. Data are presented as mean±SD. NS indicates nonsignificant.

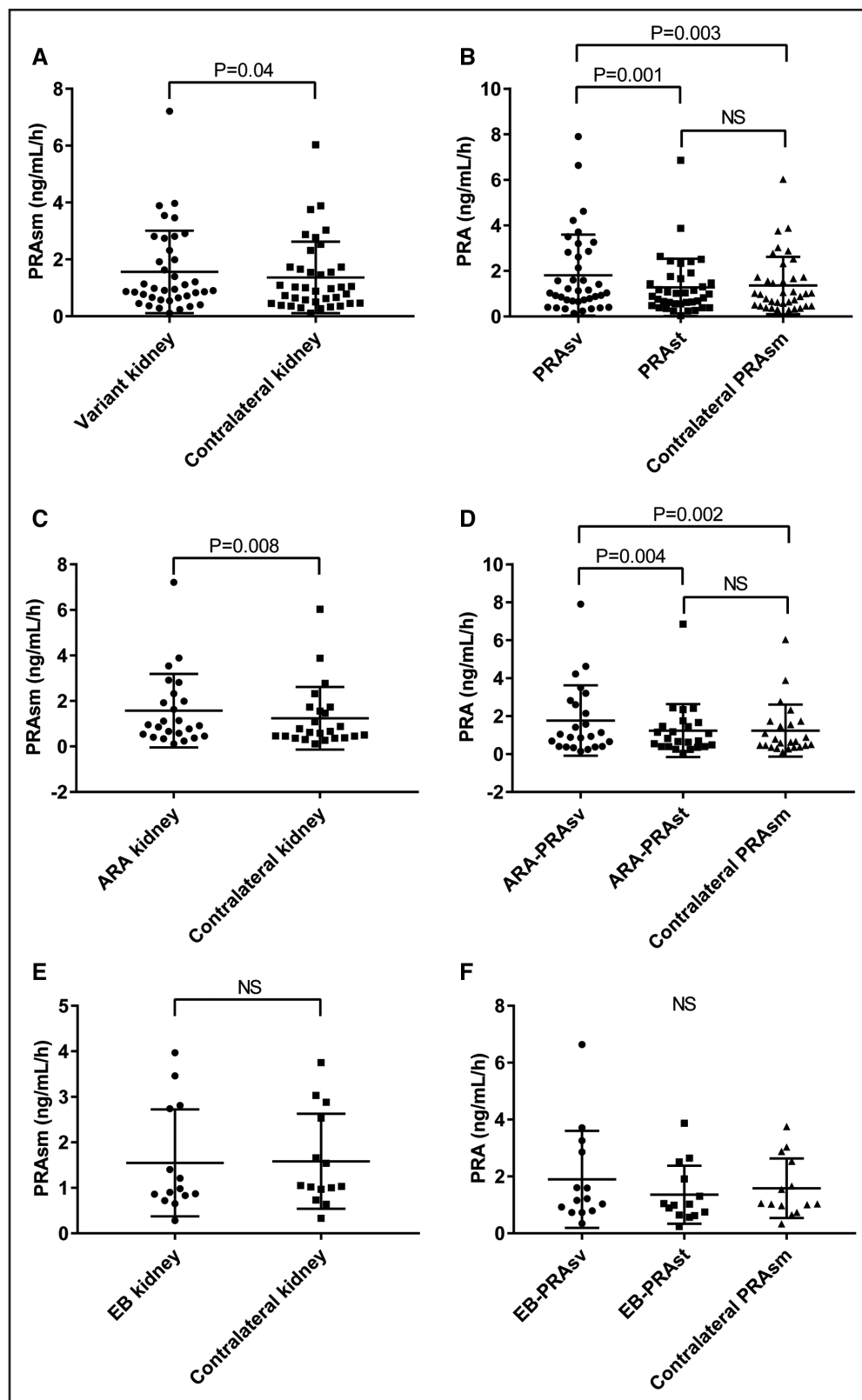


Figure 4. Comparisons of renal vein plasma renin activity (PRA) across different sampling sites in patients with unilateral solitary renal artery variation.

PRA from renal segments supplied by the variant renal artery (PRA_{sv}), PRA from renal segments supplied by the main trunk renal artery (PRA_{st}), and mean segmental renal vein PRA (PRA_{sm}) denote renin activity at respective sampling sites. **A**, **C**, and **E** show comparisons of PRA_{sm} between the variant kidney and contralateral kidney in the overall cohort, accessory renal artery (ARA) subgroup, and early branching (EB) subgroup, respectively. **B**, **D**, and **F** show comparisons among PRA_{sv}, PRA_{st}, and contralateral PRA_{sm} in the overall cohort, ARA subgroup, and EB subgroup, respectively. Data are presented as mean±SD. NS indicates nonsignificant.

Correlations of Variant Artery Hemodynamics With RAAS Markers, Anatomic Dimensions, and Clinical Parameters

Among all enrolled patients, variant artery SPG was positively correlated with ipsilateral PRA_{sm} ($r=0.36$, $P=0.03$), PRA_t ($r=0.39$, $P=0.02$), and both basal ($r=0.46$, $P=0.003$) and stimulated peripheral angiotensin II levels ($r=0.38$, $P=0.02$). After multivariable adjustment for age, sex, eGFR, UACR, 24-hour MAP and left ventricular ejection fraction, variant artery SPG remained positively associated with ipsilateral PRA_{sm} ($B=0.15$, 95% CI, 0.03–0.27; $\beta=0.43$, $P=0.02$), PRA_t ($B=0.15$ [95% CI, 0.04–0.27]; $\beta=0.44$, $P=0.01$), and both basal ($B=1.60$ [95% CI, 0.59–2.62]; $\beta=0.52$, $P=0.003$) and stimulated angiotensin II ($B=3.48$ [95% CI, 1.03–5.93]; $\beta=0.47$, $P=0.007$). In addition, variant artery SPG was positively associated with PRA_{sv} ($B=0.18$ [95% CI, 0.02–0.33]; $\beta=0.41$, $P=0.02$), PRA from renal segments supplied by main trunk renal artery ($B=0.11$ [95% CI, 0.02–0.21]; $\beta=0.37$, $P=0.02$) and peripheral basal PRA ($B=0.11$ [95% CI, 0.01–0.21]; $\beta=0.37$, $P=0.04$) after full multivariable adjustment (Table 2), supporting an exploratory hypothesis that the hemodynamic burden of anatomic variants is linked to local and systemic RAAS activation. In contrast, neither MPG nor DPG of the variant artery showed significant correlations with RAAS markers.

The variant artery was significantly narrower than the ipsilateral main artery (2.8 ± 0.8 versus 4.6 ± 0.9 mm, $P<0.001$), which was in turn narrower than the contralateral renal artery (5.0 ± 0.9 mm, $P=0.009$). Variant artery diameter was comparable between ARA and EB (2.8 ± 0.7 versus 2.8 ± 1.0 mm, $P=0.95$). In the ARA subgroup, the accessory-to-main renal artery diameter ratio correlated negatively with the DPG of the ARA ($r=-0.51$, $P=0.01$), indicating that relative hypoplasia exacerbates hemodynamic burden. In contrast, no significant association was

observed between variant artery diameter and PG in the EB subgroup.

The average nighttime MAP was positively correlated with the SPG of variant artery in the overall cohort ($r=0.34$, $P=0.04$), and this exploratory association remained evident after multivariable adjustment for age, sex, eGFR, UACR, and left ventricular ejection fraction ($B=1.68$ [95% CI, 0.62–2.75]; $\beta=0.48$, $P=0.003$; Table 2). Although a normal dipping profile ($\geq 10\%$ nocturnal systolic BP reduction) was present in only 28.9% (11/38) of patients, variant artery SPG did not significantly associate with dipping scores ($r=-0.04$, $P=0.83$). Variant artery DPG and MPG were positively correlated with eGFR ($r=0.36$, $P=0.02$ and $r=0.40$, $P=0.01$) and left ventricular ejection fraction ($r=0.46$, $P=0.004$ and $r=0.34$, $P=0.03$), whereas MPG was negatively correlated with left ventricular end-diastolic diameter ($r=-0.34$, $P=0.04$). However, these associations were no longer statistically significant after multivariable adjustment for age, sex, UACR, and 24-hour MAP, suggesting that the relationships were largely confounded by the included clinical factors.

Although renal vein and peripheral PRA exhibited no correlations with cardiorenal biomarkers, aldosterone levels were significantly associated with disease severity. Specifically, basal aldosterone correlated with urinary protein ($r=0.41$, $P=0.01$) and UACR ($r=0.39$, $P=0.01$), whereas stimulated aldosterone exhibited broader associations with urinary protein ($r=0.45$, $P=0.004$), UACR ($r=0.42$, $P=0.01$), eGFR ($r=-0.38$, $P=0.03$), and NT-proBNP ($r=0.40$, $P=0.01$). Notably, the association between eGFR and stimulated aldosterone was maintained ($B=-0.07$ [95% CI, -0.12 to -0.02]; $\beta=-0.53$, $P=0.007$) after multivariable adjustment for age, sex, UACR and 24-hour MAP, as was the correlation between urinary protein and basal aldosterone ($B=5.97$ [95% CI, 0.74–11.2]; $\beta=0.35$, $P=0.03$) after controlling for age, sex, eGFR and 24-hour MAP,

Table 2. Multivariable Linear Regression Analysis for Factors Associated With the Variant Artery SPG

Variable	B	SE	β	95% CI	P value
PRA_{sv} , ng/(mL·h)	0.18	0.08	0.41	0.02–0.33	0.02
PRA_{st} , ng/(mL·h)	0.11	0.05	0.37	0.02–0.21	0.02
Ipsilateral PRA_{sm} , ng/(mL·h)	0.15	0.06	0.43	0.03–0.27	0.02
Ipsilateral PRA_t , ng/(mL·h)	0.15	0.06	0.44	0.04–0.27	0.01
Basal PRA , ng/(mL·h)	0.11	0.05	0.37	0.01–0.21	0.04
Basal angiotensin II, pg/mL	1.60	0.50	0.52	0.59–2.62	0.003
Stimulated angiotensin II, pg/mL	3.48	1.20	0.47	1.03–5.93	0.007
Nighttime MAP, mmHg	1.68	0.52	0.48	0.62–2.75	0.003

For all variables except nighttime MAP, the model was adjusted for age, sex, eGFR, UACR, LVEF, and 24 h MAP; for nighttime MAP, the model was adjusted for age, sex, eGFR, UACR, and LVEF (excluding 24 h MAP). All variables included in the table were independent factors analyzed for their association with the variant artery SPG. β indicates standardized regression coefficient; B, unstandardized regression coefficient; eGFR, estimated glomerular filtration rate; LVEF, left ventricular ejection fraction; MAP, mean arterial pressure; PRA, plasma renin activity; PRA_{sm} , mean segmental renal vein PRA; PRA_{st} , PRA from renal segments supplied by the main trunk renal artery; PRA_{sv} , PRA from renal segments supplied by variant renal artery; PRA_t , PRA from renal vein trunk; SPG, systolic pressure gradient; and UACR, urinary albumin-to-creatinine ratio.

suggesting a potential link between aldosterone and renal damage.

Two-Year Follow-Up Outcomes

Of 38 enrolled patients, 35 (92%) completed 2-year follow-up (ARA: $n=23$; EB: $n=12$). The 3 patients lost to follow-up were evenly distributed (ARA: $n=1$; EB: $n=2$; $P=0.54$). Baseline demographic, laboratory, and intrarenal hemodynamic parameters were similar between follow-up completers and noncompleters, except for prior α -adrenergic blocker use (Table S1), suggesting minimal attrition bias. Cross-sectional comparisons at 2-year follow-up revealed no significant between-group differences in eGFR, UACR, time-averaged BP, or antihypertensive medication intensity (Table S2). Patients with unilateral ARA exhibited a significant eGFR decline from baseline to 2-year follow-up (-5.2 ± 10.1 mL/[min $\cdot$ 1.73 m 2], $P=0.02$), whereas those with EB showed no significant change (1.6 ± 7.3 mL/[min $\cdot$ 1.73 m 2], $P=0.47$; Figure S2). However, the group-by-time interaction did not achieve statistical significance ($\beta=-3.23$ [95% CI, -6.56 to 0.11]; $P=0.06$). Although the numerical difference in annual eGFR slopes (-2.34 versus $+0.89$ mL/[min $\cdot$ 1.73 m 2] per year for ARA and EB, respectively) and the observed within-group deterioration in ARA hint at a potential association with higher renal progression risk, this preliminary finding must be interpreted with caution and requires robust validation in larger, fully controlled cohorts. Point-wise least-squares means of eGFR at each visit are illustrated in Figure S3. At 2-year follow-up, UACR was significantly decreased compared with baseline in the ARA subgroup ($P=0.01$), whereas no significant decline was observed in the EB subgroup ($P=0.12$; Figure S2). Only 2 patients (5.7%) experienced proteinuria progression, with 1 case in each subgroup. No significant associations were observed between baseline hemodynamic or RAAS parameters and 2-year renal outcomes. Specifically, variant renal artery PGs (SPG, MPG, and DPG), ipsilateral renal vein PRA (PRA_{sv}, PRA_{sm}, and PRA_l), and peripheral RAAS markers (renin, angiotensin II, and aldosterone) showed no correlation with either eGFR change or UACR change over 24 months.

DISCUSSION

Beyond renal artery stenosis, renal artery variations attract growing interest, particularly in individuals with hypertension and target-organ damage.^{14,23} To the best of our knowledge, this study represents the first systematic application of pressure wire–based hemodynamic assessment and selective segmental renal vein renin sampling to evaluate renal artery variations in hypertensive CKD. Our principal findings are 4-fold: (1) renal artery variations cause measurable intrarenal hemodynamic alterations, with functional impairment observed

primarily in ARA; (2) these hemodynamic changes are associated with localized renin hypersecretion; (3) SPG showed an exploratory positive association with both localized and systemic RAAS activation, which potentially correlates with further target-organ impairment; and (4) patients with ARA, but not EB, showed a preliminary trend toward a 2-year renal function decline, requiring future validation.

Prior studies of renal artery variations were largely limited to anatomic description and observational associations with hypertension.^{13,24} Our findings advance this field by demonstrating that these variants cause quantifiable hemodynamic impairment comparable in mechanism to renal artery stenosis, albeit at a milder magnitude. Although the observed PGs (eg, MPG 1.10 ± 3.39 mmHg) were substantially below hemodynamically significant thresholds for revascularization in renal artery stenosis (≥ 10 mmHg),²⁵ both MPG and DPG were higher in ARAs than in ipsilateral main and contralateral renal arteries. This indicates subtle, localized hemodynamic disturbances. Anatomically, ARAs are typically narrower and longer than normal renal arteries, supplying 20% to 40% of total renal blood flow independently.^{26–28} According to Poiseuille law, this geometry creates a chronically high-resistance, low-perfusion microenvironment susceptible to mild pressure drops. Animal models confirm that even a minor 5% reduction in renal perfusion pressure decreases systolic flow, whereas a 25% reduction upregulates renin secretion.²⁹ This uneven intrarenal pressure distribution may cause hypoperfusion in ARA-supplied segments, leading to localized renin hypersecretion associated with SPG, aligning with previous reports and providing preliminary biological evidence that these seemingly subclinical microgradients may be capable of activating the RAAS.³⁰

Although previous studies have reported elevated peripheral PRA in ARA patients,^{11,12} our selective segmental sampling precisely localizes this renin hypersecretion to the specific renal segments supplied by the ARA. Crucially, this segmental renin lateralization occurs with preserved main trunk PRA. Traditional reliance on main renal vein sampling would fail to detect this pathophysiology, as renin signals from affected segments are diluted by blood from normally perfused renal parenchyma. This methodological insight highlights the value of selective segmental sampling and provides proof-of-concept rationale for reclassification of anatomic variants from benign to potentially functional, analogous to the established concept in renovascular hypertension.^{31,32}

Higher variant artery SPG showed an exploratory association with elevated renin activity in peripheral circulation, and in both segmental and trunk veins of the ipsilateral kidney—suggesting that hemodynamic burden may contribute to both local and systemic RAAS activation. Similar links are well established in renal artery stenosis.^{17,33,34} In addition, the ARA-to-main renal

artery diameter ratio was negatively correlated with ARA-induced DPG. Unlike MPG or DPG—which represent steady-state and resistance-related perfusion pressure^{35–37}—SPG reflects the pulsatile, peak-to-peak hemodynamic burden imposed on the renal vascular bed and correlates with peak systolic velocity.³⁸ Intriguingly, SPG was comparable across all studied arteries, whereas MPG and DPG were significantly higher in the ARA. This suggests that SPG may reflect the sensitivity of juxtaglomerular cells to pulsatile stress via baroreceptor mechanism,^{39–41} consistent with its observed association with nighttime MAP. However, SPG did not associate with nighttime dipping status. This dissociation reflects that high baseline circadian disruption likely renders dipping status insensitive to local hemodynamic factors in CKD.⁴² Consequently, variant-related stress elevates absolute nocturnal pressure via RAAS activation without altering already disrupted circadian rhythms. In clinical application, SPG may serve as a functional indicator for renin hypersecretion, whereas MPG and DPG act as structural markers for detecting subtle hemodynamic differences of renal artery variations.

The dissociation between renin and aldosterone patterns in our study is noteworthy. While segmental renin activity correlated with variant-induced SPG, peripheral aldosterone did not, likely due to systemic dilution and counter-regulation by unaffected segments. In hypertensive CKD, circulating aldosterone is heavily modulated by systemic factors, including volume overload, sympathetic overactivation, and impaired clearance.^{43–45} Rather than reflecting localized hemodynamics, aldosterone showed closer associations with systemic markers of target-organ injury, including proteinuria, eGFR, and NT-proBNP. Excess aldosterone promotes cardiorenal decline through genomic and nongenomic pathways that drive inflammation, fibrosis, and podocyte injury.⁴⁶ Consequently, targeting aldosterone with a nonsteroidal mineralocorticoid receptor antagonist may represent a potential therapeutic strategy to mitigate CKD progression and reduce cardiovascular complications in this specific patient subgroup.⁴⁷

Although systemic RAAS inhibition reduces efferent arteriolar tone to exert antiproteinuric effects in the ARA group, this functional relief may not fully counteract the variant-induced localized fixed resistance, potentially accelerating tubulointerstitial fibrosis and nephron loss. Notably, within-group analysis revealed a significant decrease in eGFR within the ARA group over 2 years, whereas eGFR remained statistically unchanged in EB patients—who characteristically maintain a favorable total vascular cross-sectional area and homogeneous intrarenal perfusion.⁴⁸ However, the lack of a significant group-by-time interaction precludes definitive inferences regarding differential renal progression between ARA and EB, and these preliminary trends should be interpreted with extreme caution and regarded as

hypothesis-generating. Importantly, baseline hemodynamic and RAAS parameters were not associated with 2-year changes in eGFR or UACR. This negative finding tempers the clinical relevance of our mechanistic insights, suggesting that local hemodynamic stress and acute RAAS activation may not directly drive mid-term renal decline. Although these results constitute proof-of-concept mechanistic observations requiring validation in larger longitudinal cohorts, they nevertheless challenge the traditional view of renal artery variants as merely incidental findings. Rather, our data support a potential functional reclassification based on hemodynamic assessment, highlighting that patients with ARAs, high PGs, or segmental renin lateralization may represent a unique microhemodynamic phenotype—a concept that carries potential implications for long-term functional monitoring in future prospective cohorts.

The selection of patients with hypertensive CKD was dictated by both pathophysiological and clinical considerations. In this high-risk population, ongoing nephron loss and parenchymal deterioration often intertwine with secondary hypertensive pathways, making the evaluation of variant-related micro-hemodynamic strain highly relevant. Furthermore, established renal impairment clinically justifies invasive functional assessment, which would be ethically unwarranted in uncomplicated hypertension. Mechanistically, this core cascade may extrapolate to non-CKD hypertensives because the PGs and subsequent segmental RAAS activation are fundamentally dictated by vascular geometry. However, individuals with preserved renal function possess greater compensatory reserve and may tolerate mild hypoperfusion subclinically without overt systemic impairment. Whether these variant-related alterations represent a preclinical stage leading to future parenchymal disease requires prospective validation in non-CKD cohorts.

Several limitations of this pilot study warrant consideration. First, the methodological framework is constrained by a single-center design, a modest sample size, and a relatively short 2-year observation period. Although intra-individual comparisons enhanced statistical efficiency, the limited cohort size reduces statistical power, constrains linear mixed-effects model structural flexibility, prevents identification of definitive hemodynamic thresholds, and risks potential overfitting that destabilizes multivariable models, rendering these findings strictly exploratory. Second, repeat ambulatory BP monitoring was not obtained during follow-up, limiting our ability to evaluate temporal variations in circadian rhythms. Third, all hemodynamic and hormonal assessments were single time point measurements; notably, the 2-week washout may not completely eliminate heterogeneous residual renin activity from prior ACE inhibitors/angiotensin II receptor blockers therapy. Consequently, these measurements could partially reflect shared interindividual or withdrawal-related variability rather than exclusive local mechanisms. However, we

attempted to mitigate this confounding by analyzing relative differences, and this localized link is further supported by renin hypersecretion being confined to ARA-perfused segments. Fourth, the intraoperative phase was unblinded due to the requirement for real-time visual tracking during angiography, although gradient calculations were performed by independent investigators to minimize bias. In conclusion, whereas ethical and safety concerns preclude repeated invasive evaluations, future multicenter, longitudinal trials incorporating serial ambulatory BP monitoring and noninvasive imaging modalities are required to validate these hemodynamic alterations and evaluate whether tailored management strategies can optimize long-term clinical outcomes.

PERSPECTIVES

This study provides proof-of-concept evidence that renal artery variations, particularly ARAs, may represent functionally active variants associated with localized RAAS activation in hypertensive CKD. Pressure wire-based hemodynamic assessment combined with selective segmental renin sampling offers a potential framework for identifying specific micro-hemodynamic phenotypes that might justify longitudinal observation. Transitioning from anatomic observation to physiological phenotyping highlights a potential clinical implication for cardiorenal risk evaluation. Future prospective, longitudinal studies are warranted to validate these hemodynamic alterations and evaluate whether targeted management strategies can improve clinical outcomes. Ultimately, these exploratory insights lay the groundwork for refining future risk-stratification strategies for patients with renal artery variants.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Tables S1–S2
Figures S1–S3
STROBE Checklist

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